```
from google.colab import drive
drive.mount('/content/drive', force_remount=True)

import os
os.chdir('/content/drive/My Drive/Math586')
base_directory = os.getcwd()
datas = base_directory + '/data/'
```

```
Mounted at /content/drive
```

```
import matplotlib.pyplot as plt

styledict = {'xtick.labelsize':20,
             'xtick.major.size':9,
             'xtick.major.width':1,
             'ytick.labelsize':20,
             'ytick.major.size':9,
             'ytick.major.width':1,
             'legend.framealpha':0.0,
             'legend.fontsize':15,
             'axes.labelsize':20,
             'axes.titlesize':25,
             'axes.linewidth':2,
             'figure.figsize':(12,8),
             'savefig.format':'jpg'}
plt.rcParams.update(**styledict)

import numpy as np
from matplotlib import cm as cm1
from matplotlib.colors import ListedColormap, to_rgba

def newColMap(colors):
    """
    This function creates a new color map from a list of colors given
    as a parameter. Recommended length of list of colors is at least
6.
    """
    first = np.repeat([to_rgba(colors[0])], 2, axis = 0)
    last = np.repeat([to_rgba(colors[-1])], 2, axis = 0)
    v = cm1.get_cmap('viridis', 16*(len(colors)-2))
    newcolors = v(np.linspace(0, 1, 16*(len(colors)-2)))
    for (i, col) in enumerate(colors[1:-1]):
        newcolors[16*i : 16*(i+1), :] = to_rgba(col)
    return ListedColormap(np.append(np.append(first,newcolors,
axis=0), last, axis=0))

import numpy as np
import matplotlib.pyplot as plt
from matplotlib.colors import rgb2hex
from matplotlib.patches import Polygon
```

```
import matplotlib.mlab as mlab
import pandas as pd

#from cartopy import crs, mpl
#import matplotlib.ticker as mticker
#from cartopy.mpl.gridliner import LONGITUDE_FORMATTER,
LATITUDE_FORMATTER
#from mpl_toolkits import mplot3d
#import netCDF4
#from netCDF4 import Dataset as ds
from urllib import request
import scipy as sp
from scipy import stats as stt
from sklearn import datasets, linear_model
from sklearn.neighbors import KernelDensity as kd
from sklearn.linear_model import LinearRegression
from random import randint
from scipy.stats import norm
import statsmodels
import sklearn
import math
import statistics
import sympy as sy
from sympy import symbols, diff
import statsmodels.api as sm
from datetime import date
```

Now The Actual Code with the 3 Califronia Cities

```
Eureka = datas + "EurekaNorCal42to25.csv" #1942-2025
SanFrancisco = datas + "SFYearly45to25.csv" #1945-2025
SanDiego = datas + "SDYearly43to25.csv" #1943-2025
EU = pd.read_csv(Eureka)
SF = pd.read_csv(SanFrancisco)
SD = pd.read_csv(SanDiego)

SD
column_names = EU.columns.tolist()
print(column_names)
#Column Names of Importance:
#'PRCP' #54 for SD, 62 for EU and SF
#'TAVG' #60 for SD, 66 for SF, 68 for EU
#'TMAX' #62 for SD, 68 for SF, 70 for EU
#Start all at around 1951
```

```
['STATION', 'DATE', 'LATITUDE', 'LONGITUDE', 'ELEVATION', 'NAME',
'AWND', 'AWND_ATTRIBUTES', 'CDSD', 'CDSD_ATTRIBUTES', 'CLDD',
'CLDD_ATTRIBUTES', 'DP01', 'DP01_ATTRIBUTES', 'DP10',
'DP10_ATTRIBUTES', 'DP1X', 'DP1X_ATTRIBUTES', 'DSND',
'DSND_ATTRIBUTES', 'DSNW', 'DSNW_ATTRIBUTES', 'DT00',
'DT00_ATTRIBUTES', 'DT32', 'DT32_ATTRIBUTES', 'DX32',
'DX32_ATTRIBUTES', 'DX70', 'DX70_ATTRIBUTES', 'DX90',
'DX90_ATTRIBUTES', 'DYFG', 'DYFG_ATTRIBUTES', 'DYHF',
'DYHF_ATTRIBUTES', 'DYTS', 'DYTS_ATTRIBUTES', 'EMNT',
'EMNT_ATTRIBUTES', 'EMSD', 'EMSD_ATTRIBUTES', 'EMSN',
'EMSN_ATTRIBUTES', 'EMXP', 'EMXP_ATTRIBUTES', 'EMXT',
'EMXT_ATTRIBUTES', 'FZF0', 'FZF0_ATTRIBUTES', 'FZF1',
'FZF1_ATTRIBUTES', 'FZF2', 'FZF2_ATTRIBUTES', 'FZF5',
'FZF5_ATTRIBUTES', 'FZF6', 'FZF6_ATTRIBUTES', 'HDSD',
'HDSD_ATTRIBUTES', 'HTDD', 'HTDD_ATTRIBUTES', 'PRCP',
'PRCP_ATTRIBUTES', 'PSUN', 'PSUN_ATTRIBUTES', 'SNOW',
'SNOW_ATTRIBUTES', 'TAVG', 'TAVG_ATTRIBUTES', 'TMAX',
'TMAX_ATTRIBUTES', 'TMIN', 'TMIN_ATTRIBUTES', 'TSUN',
'TSUN_ATTRIBUTES', 'WDF2', 'WDF2_ATTRIBUTES', 'WDF5',
'WDF5_ATTRIBUTES', 'WDFG', 'WDFG_ATTRIBUTES', 'WDFM',
'WDFM_ATTRIBUTES', 'WSF2', 'WSF2_ATTRIBUTES', 'WSF5',
'WSF5_ATTRIBUTES', 'WSFG', 'WSFG_ATTRIBUTES', 'WSFM',
'WSFM_ATTRIBUTES']

SF.iloc[:,66]
#SF
```

Getting the important data matrices. Average temperature, Average Max Temperature, and Precipitation totals

```
EUprcp = EU.iloc[9:83,62].values #1951-2024 Yearly Precipitation Data
of Eureka California
EUtavg = EU.iloc[9:83,68].values # 1951-2024 Yearly Temp Avg Data of
Eureka California
EUtmax = EU.iloc[9:83,70].values #1951-2024 Yearly Temp Max Data of
Eureka California
SFprcp = SF.iloc[6:80,62].values #1951-2024 Yearly Precipitation Data
of San Francisco California
SFtavg = SF.iloc[6:80,66].values # 1951-2024 Yearly Temp Avg Data of
San Fran California
SFtmax = SF.iloc[6:80,68].values # 1951-2024 Yearly Temp Max Data of
San Fran California
SDprcp = SD.iloc[12:86,54].values # 1951-2024 Yearly Precipitation
Data of San Diego California
SDtavg = SD.iloc[12:86,60].values # 1951-2024 Yearly Temp Avg Data of
San Diego California
SDtmax = SD.iloc[12:86,62].values # 1951-2024 Yearly Temp Max Data of
San Diego California
```

These are the Functions for Trends and Trend plots

Essentially, just showing the change from 1951 to 2024

```
def TAVG_Trend(data, start, end):
    subset = data[(data['DATE'] >= start) & (data['DATE'] <= end)]

    subset = subset.dropna(subset=['DATE', 'TAVG'])


    x = subset['DATE'].values.reshape(-1, 1)
    y = subset['TAVG'].values.reshape(-1, 1)

    model = LinearRegression().fit(x, y)
    slope = model.coef_[0][0]  # Slope of the linear regression model

    return slope

def TMAX_Trend(data, start, end):
    subset = data[(data['DATE'] >= start) & (data['DATE'] <= end)]

    subset = subset.dropna(subset=['DATE', 'TMAX'])


    x = subset['DATE'].values.reshape(-1, 1)
    y = subset['TMAX'].values.reshape(-1, 1)

    model = LinearRegression().fit(x, y)
    slope = model.coef_[0][0]  # Slope of the linear regression model

    return slope

def PRCP_Trend(data, start, end):
    subset = data[(data['DATE'] >= start) & (data['DATE'] <= end)]

    subset = subset.dropna(subset=['DATE', 'PRCP'])


    x = subset['DATE'].values.reshape(-1, 1)
    y = subset['PRCP'].values.reshape(-1, 1)

    model = LinearRegression().fit(x, y)
    slope = model.coef_[0][0]  # Slope of the linear regression model

    return slope

def TAVG_ltl(data, start_year, end_year, color):
    subset = data[(data['DATE'] >= start_year) & (data['DATE'] <=
```

```
end_year)]

    subset = subset.dropna(subset=['DATE', 'TAVG'])

    x = subset['DATE'].values.reshape(-1, 1)
    y = subset['TAVG'].values.reshape(-1, 1)

    model = LinearRegression().fit(x, y)
    trend_line = model.predict(x)

    slope = model.coef_[0][0]

    plt.plot(subset['DATE'], trend_line, label=f"Linear Trend from
"f'{start_year}-{end_year}', color=color, linestyle='-')
    plt.annotate(f"Trend: {slope:.3f}",
             xy=(subset['DATE'].iloc[-1], trend_line[-1]),
             xytext=(-50, 100),
             textcoords='offset points',
             fontsize=10, color=color, fontweight='bold')


def TMAX_ltl(data, start_year, end_year, color):
    subset = data[(data['DATE'] >= start_year) & (data['DATE'] <=
end_year)]

    subset = subset.dropna(subset=['DATE', 'TMAX'])

    x = subset['DATE'].values.reshape(-1, 1)
    y = subset['TMAX'].values.reshape(-1, 1)

    model = LinearRegression().fit(x, y)
    trend_line = model.predict(x)

    slope = model.coef_[0][0]

    plt.plot(subset['DATE'], trend_line, label=f"Linear Trend from
"f'{start_year}-{end_year}', color=color, linestyle='-')
    plt.annotate(f"Trend: {slope:.3f}",
             xy=(subset['DATE'].iloc[-1], trend_line[-1]),
             xytext=(-50, -100),
             textcoords='offset points',
             fontsize=10, color=color, fontweight='bold')

def PRCP_ltl(data, start_year, end_year, color):
    subset = data[(data['DATE'] >= start_year) & (data['DATE'] <=
end_year)]

    subset = subset.dropna(subset=['DATE', 'PRCP'])

    x = subset['DATE'].values.reshape(-1, 1)
    y = subset['PRCP'].values.reshape(-1, 1)
```

```
    model = LinearRegression().fit(x, y)
    trend_line = model.predict(x)

    slope = model.coef_[0][0]

    plt.plot(subset['DATE'], trend_line, label=f"Linear Trend from
"f'{start_year}-{end_year}', color=color, linestyle='-')
    plt.annotate(f"Trend: {slope:.3f}",
             xy=(subset['DATE'].iloc[-1], trend_line[-1]),
             xytext=(-50, 125),
             textcoords='offset points',
             fontsize=10, color=color, fontweight='bold')
```

Yearly Average Temperature

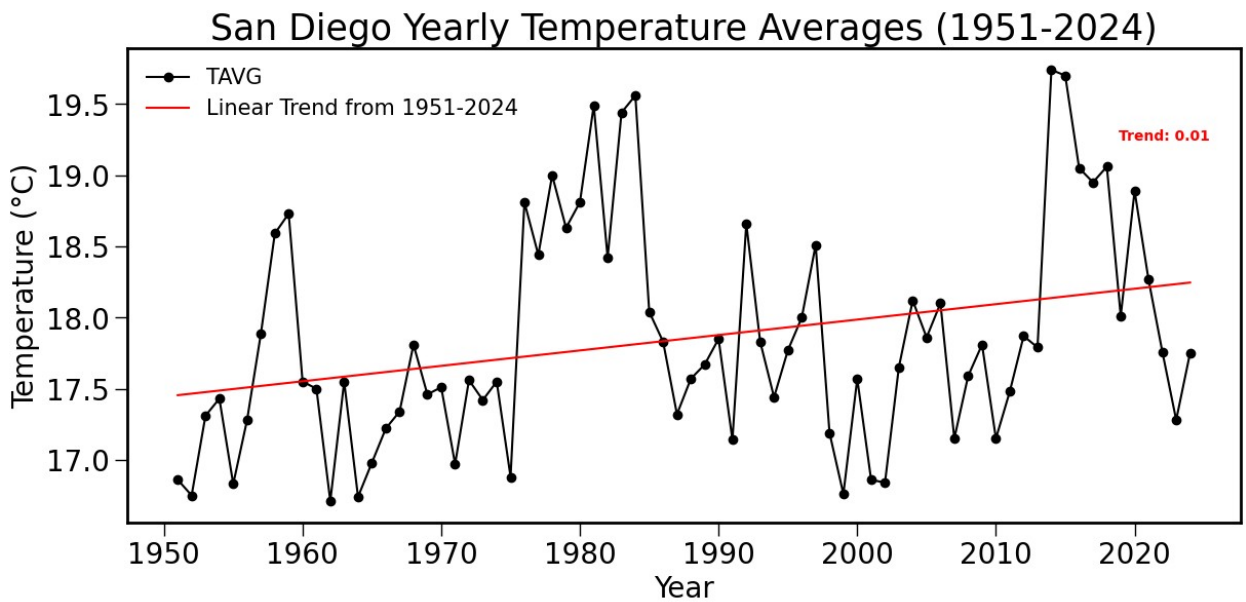
Graphs and Trends

```
SDatrends = {}
SFatrends = {}
EUatrends = {}
for start, end in [(1951, 2024)]:
    SDTavg_Trend = TAVG_Trend(SD, start, end)
    SFTavg_Trend = TAVG_Trend(SF, start, end)
    EUTavg_Trend = TAVG_Trend(EU, start, end)
    SDatrends[f'{start}-{end}'] = SDTavg_Trend
    SFatrends[f'{start}-{end}'] = SFTavg_Trend
    EUatrends[f'{start}-{end}'] = EUTavg_Trend
for period, trend in SDatrends.items():
  print(f'Temporal trend of TAVG in San Diego per decade for {period}:
{trend:.4f} °C per year')
for period, trend in SFatrends.items():
  print(f'Temporal trend of TAVG in San Francisco per decade for
{period}: {trend:.4f} °C per year')
for period, trend in EUatrends.items():
  print(f'Temporal trend of TAVG in Eureka per decade for {period}:
{trend:.4f} °C per year')
```

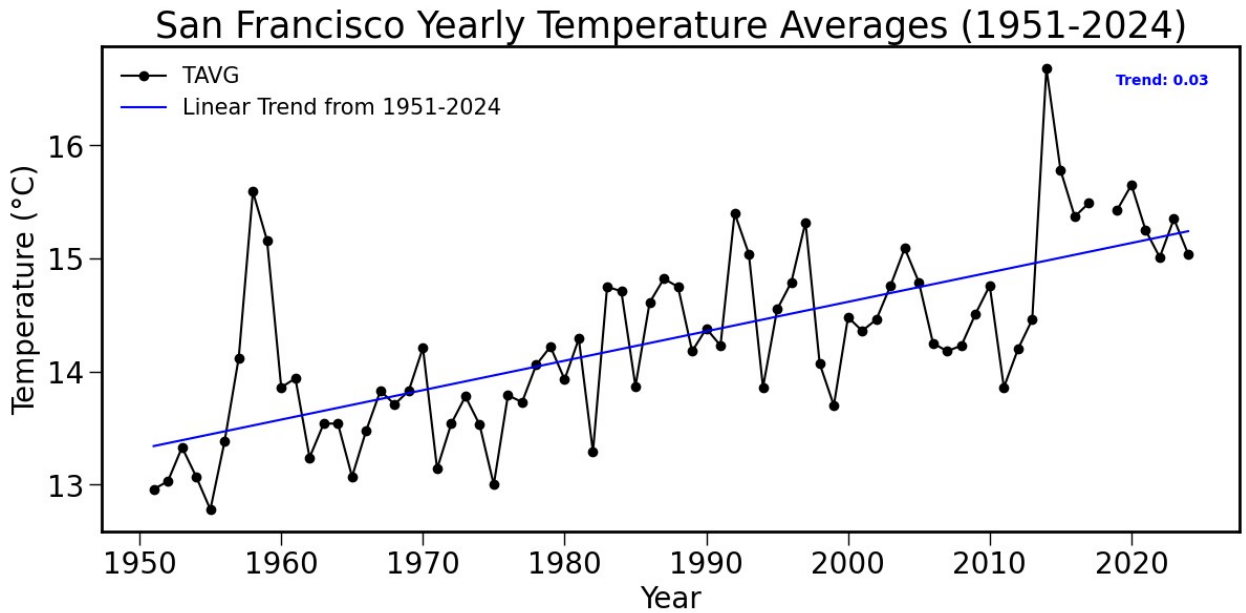
```
Temporal trend of TAVG in San Diego per decade for 1951-2024: 0.0108
°C per year
Temporal trend of TAVG in San Francisco per decade for 1951-2024:
0.0260 °C per year
Temporal trend of TAVG in Eureka per decade for 1951-2024: 0.0098 °C
per year
```

```
#SD['TAVG'] = pd.to_numeric(SD['AVG'], errors='coerce')
YEARS = np.linspace(1951, 2024, 74)
SDtavg = SDtavg
```

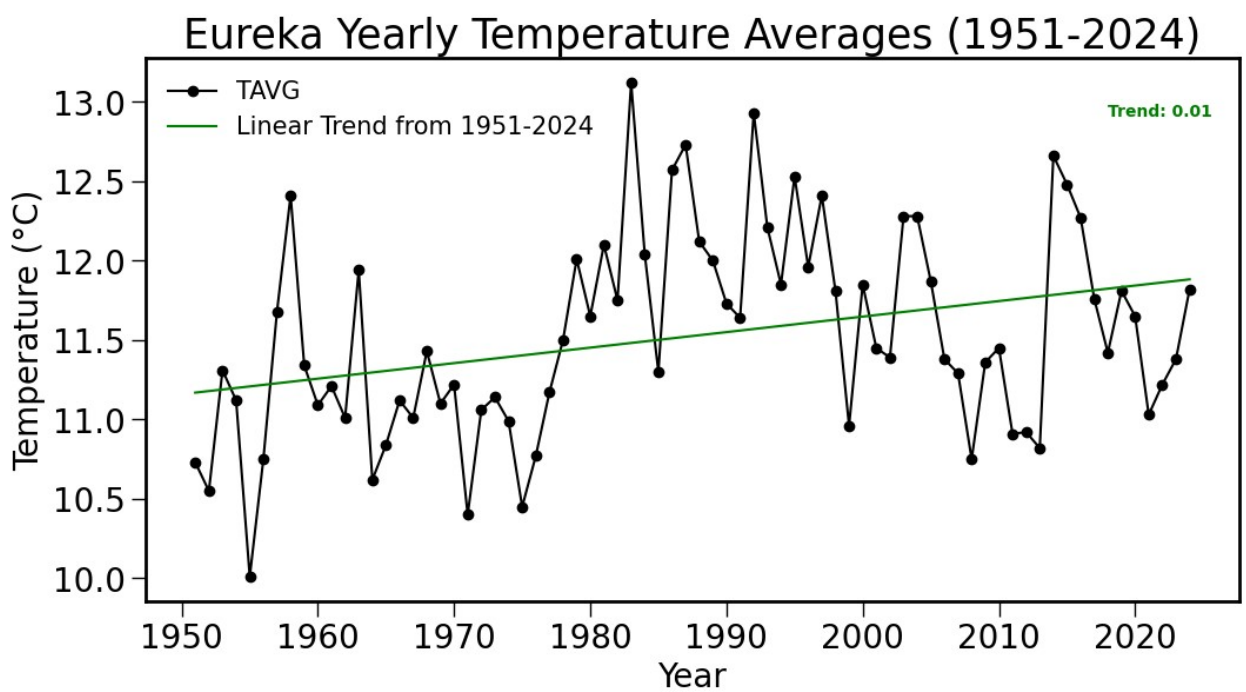
```
plt.figure(figsize=(14, 6))
plt.plot(YEARS, SDtavg, 'k-o', label='TAVG')
TAVG_ltl(SD, 1951, 2024, 'red')
plt.title('San Diego Yearly Temperature Averages (1951-2024)')
plt.xlabel('Year')
plt.ylabel('Temperature (°C)')
plt.legend()
plt.show()
```




```
YEARS = np.linspace(1951, 2024, 74)
plt.figure(figsize=(14, 6))
plt.plot(YEARS, SFtavg, 'k-o', label='TAVG')
TAVG_ltl(SF, 1951, 2024, 'blue')
plt.title('San Francisco Yearly Temperature Averages (1951-2024)')
plt.xlabel('Year')
plt.ylabel('Temperature (°C)')
plt.legend()
plt.show()
```




```
YEARS = np.linspace(1951, 2024, 74)
plt.figure(figsize=(12, 6))
plt.plot(YEARS, EUtavg, 'k-o', label='TAVG')
TAVG_ltl(EU, 1951, 2024, 'green')
plt.title('Eureka Yearly Temperature Averages (1951-2024)')
plt.xlabel('Year')
plt.ylabel('Temperature (°C)')
plt.legend()
plt.show()
```



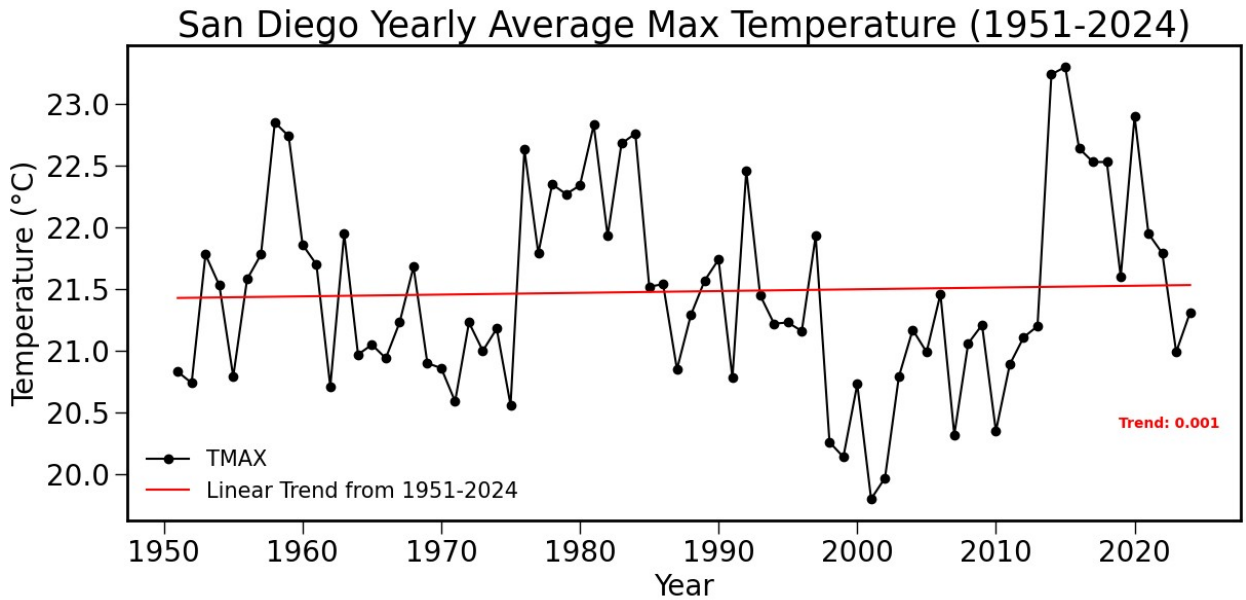
Yearly Average MAX Temperature

Graphs and Trends

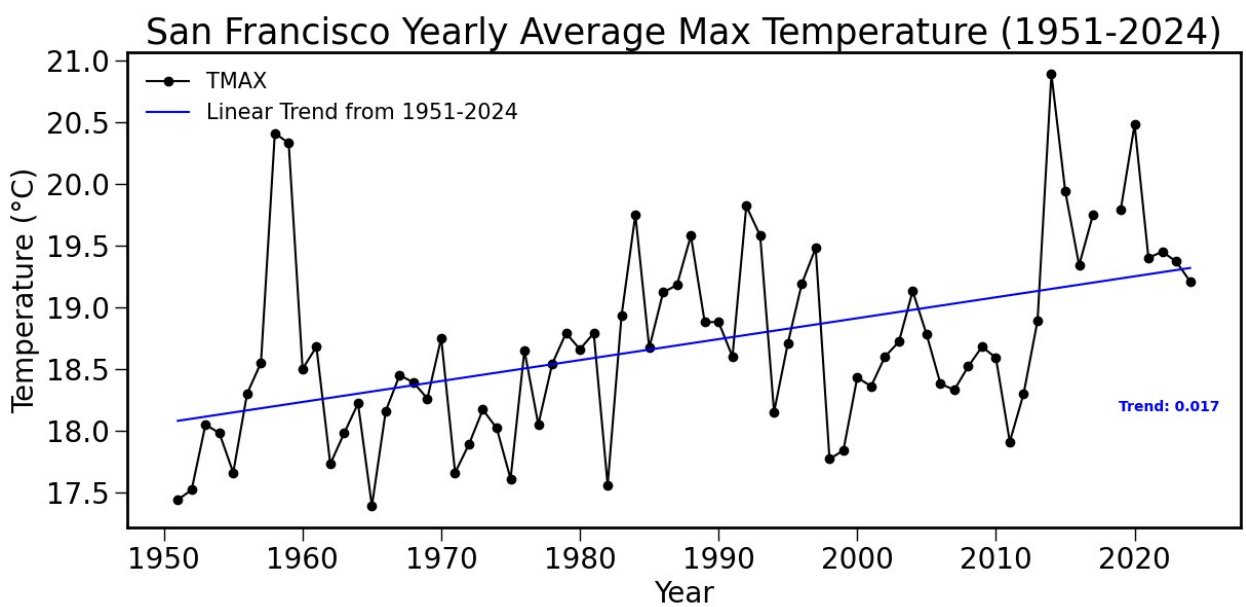
```
SDmtrends = {}
SFmtrends = {}
EUmtrends = {}
for start, end in [(1951, 2024)]:
    SDTmax_Trend = TMAX_Trend(SD, start, end)
    SFTmax_Trend = TMAX_Trend(SF, start, end)
    EUTmax_Trend = TMAX_Trend(EU, start, end)
    SDmtrends[f'{start}-{end}'] = SDTmax_Trend
    SFmtrends[f'{start}-{end}'] = SFTmax_Trend
    EUmtrends[f'{start}-{end}'] = EUTmax_Trend
for period, trend in SDmtrends.items():
  print(f'Temporal trend of TMAX in San Diego per decade for {period}:
{trend:.4f} °C per year')
for period, trend in SFmtrends.items():
  print(f'Temporal trend of TMAX in San Francisco per decade for
{period}: {trend:.4f} °C per year')
for period, trend in EUmtrends.items():
  print(f'Temporal trend of TMAX in Eureka per decade for {period}:
{trend:.4f} °C per year')
```

```
Temporal trend of TMAX in San Diego per decade for 1951-2024: 0.0014
°C per year
Temporal trend of TMAX in San Francisco per decade for 1951-2024:
0.0170 °C per year
Temporal trend of TMAX in Eureka per decade for 1951-2024: 0.0236 °C
per year
```

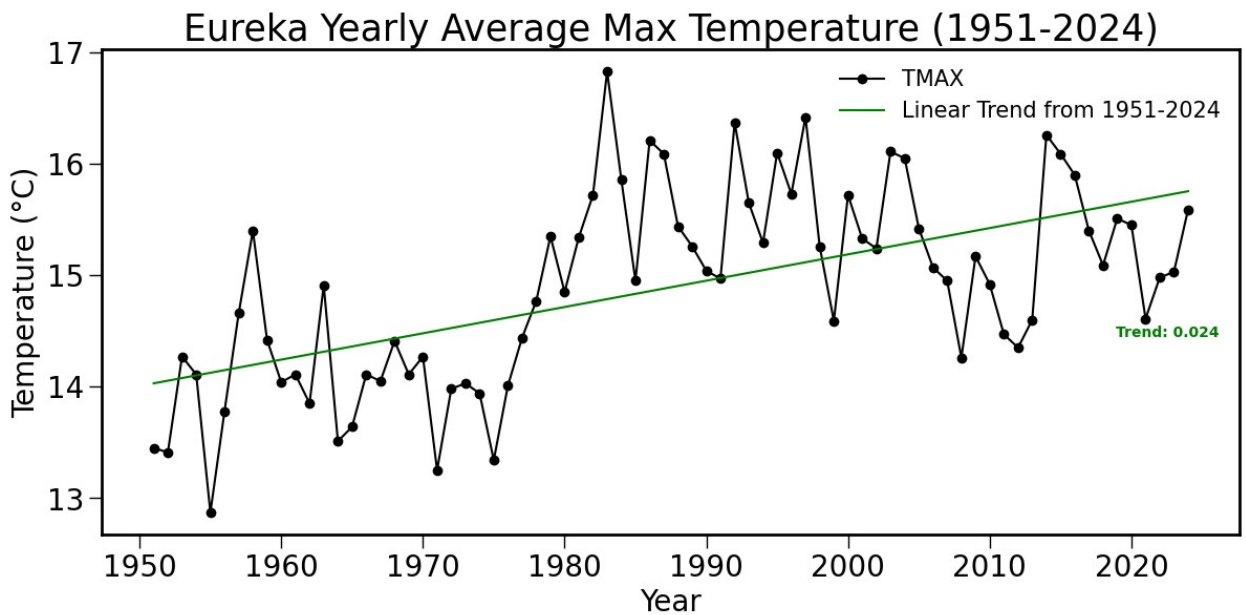
```
YEARS = np.linspace(1951, 2024, 74)
SDtmax = SDtmax
plt.figure(figsize=(14, 6))
plt.plot(YEARS, SDtmax, 'k-o', label='TMAX')
TMAX_ltl(SD, 1951, 2024, 'red')
plt.title('San Diego Yearly Average Max Temperature (1951-2024)')
plt.xlabel('Year')
plt.ylabel('Temperature (°C)')
plt.legend()
plt.show()
```




```
YEARS = np.linspace(1951, 2024, 74)
plt.figure(figsize=(14, 6))
plt.plot(YEARS, SFtmax, 'k-o', label='TMAX')
TMAX_ltl(SF, 1951, 2024, 'blue')
plt.title('San Francisco Yearly Average Max Temperature (1951-2024)')
plt.xlabel('Year')
plt.ylabel('Temperature (°C)')
plt.legend()
plt.show()
```



```
YEARS = np.linspace(1951, 2024, 74)
plt.figure(figsize=(14, 6))
plt.plot(YEARS, EUtmax, 'k-o', label='TMAX')
TMAX_ltl(EU, 1951, 2024, 'green')
plt.title('Eureka Yearly Average Max Temperature (1951-2024)')
plt.xlabel('Year')
plt.ylabel('Temperature (°C)')
plt.legend()
plt.show()
```



Yearly Precipitation Totals

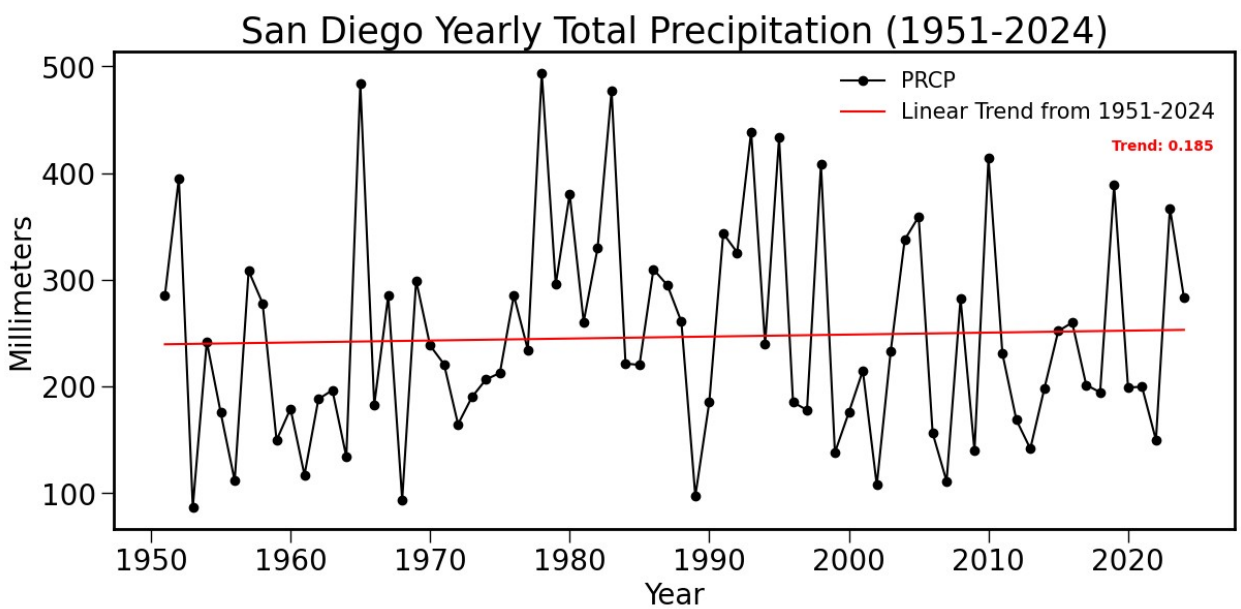
Graphs and Trends

```
SDptrends = {}
SFptrends = {}
EUptrends = {}
for start, end in [(1951, 2024)]:
    SDp_Trend = PRCP_Trend(SD, start, end)
    SFp_Trend = PRCP_Trend(SF, start, end)
    EUp_Trend = PRCP_Trend(EU, start, end)
    SDptrends[f'{start}-{end}'] = SDp_Trend
    SFptrends[f'{start}-{end}'] = SFp_Trend
    EUptrends[f'{start}-{end}'] = EUp_Trend
for period, trend in SDptrends.items():
  print(f'Temporal trend of Precipitation in San Diego per decade for
{period}: {trend:.4f} °C per year')
for period, trend in SFptrends.items():
```

```
  print(f'Temporal trend of Precipitation in San Francisco per decade
for {period}: {trend:.4f} °C per year')
for period, trend in EUptrends.items():
  print(f'Temporal trend of Precipitation in Eureka per decade for
{period}: {trend:.4f} °C per year')
```

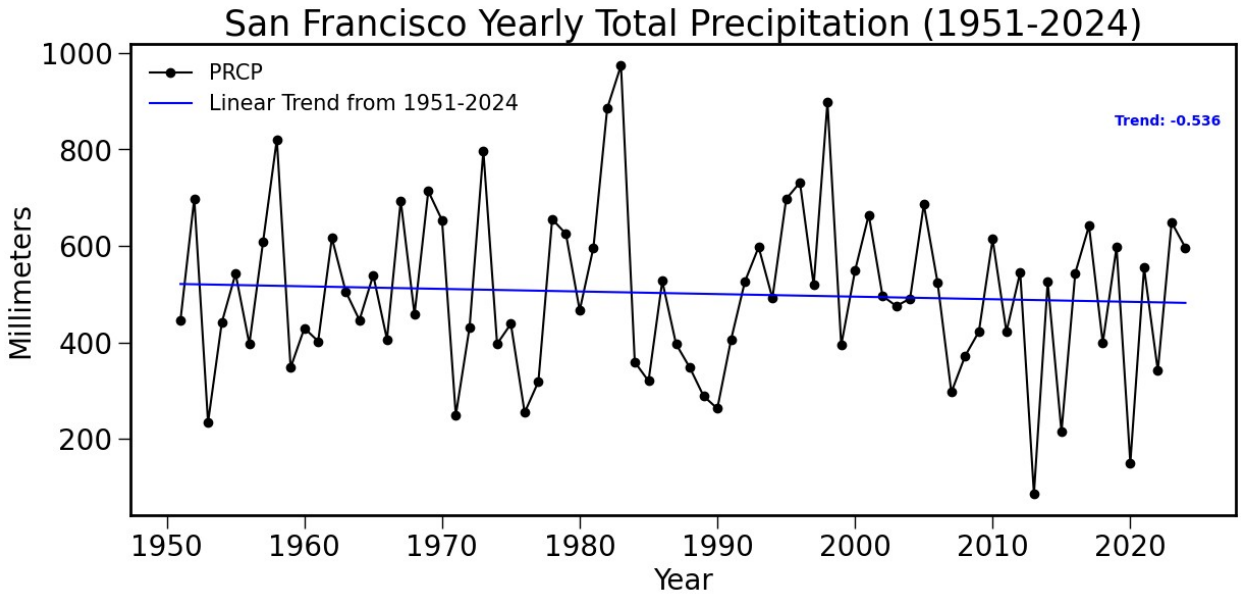
```
Temporal trend of Precipitation in San Diego per decade for 1951-2024:
0.1848 °C per year
Temporal trend of Precipitation in San Francisco per decade for 1951-
2024: -0.5356 °C per year
Temporal trend of Precipitation in Eureka per decade for 1951-2024:
0.0951 °C per year
```

```
YEARS = np.linspace(1951, 2024, 74)
plt.figure(figsize=(14, 6))
plt.plot(YEARS, SDprcp, 'k-o', label='PRCP')
PRCP_ltl(SD, 1951, 2024, 'red')
plt.title('San Diego Yearly Total Precipitation (1951-2024)')
plt.xlabel('Year')
plt.ylabel('Millimeters')
plt.legend()
plt.show()
```

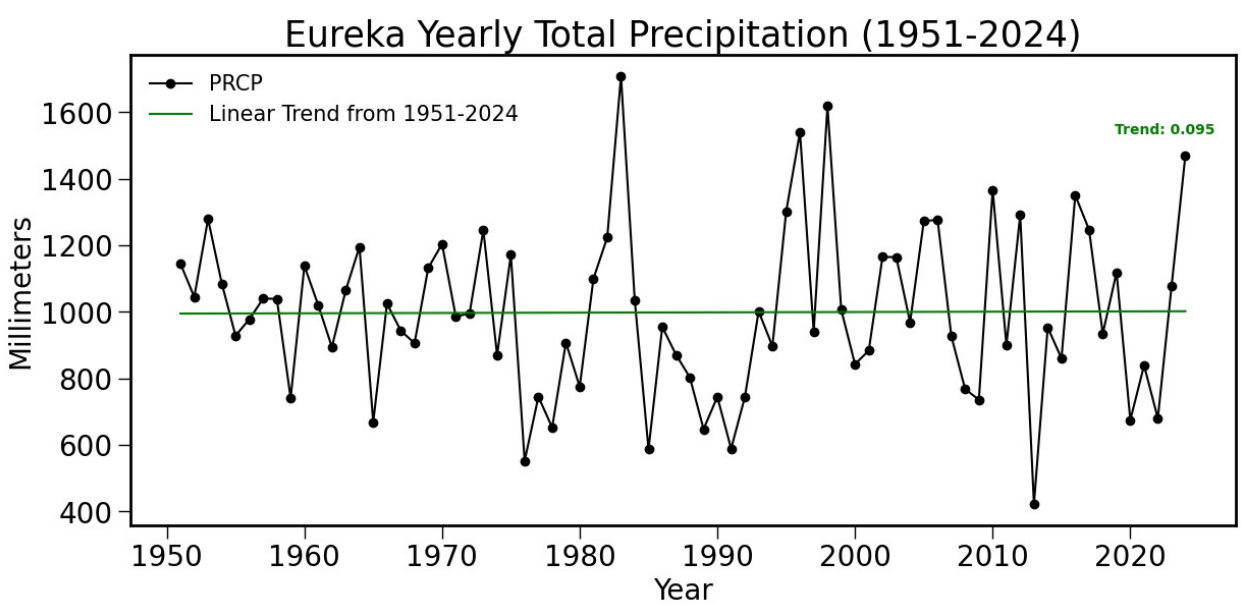



```
YEARS = np.linspace(1951, 2024, 74)
plt.figure(figsize=(14, 6))
plt.plot(YEARS, SFprcp, 'k-o', label='PRCP')
PRCP_ltl(SF, 1951, 2024, 'blue')
plt.title('San Francisco Yearly Total Precipitation (1951-2024)')
plt.xlabel('Year')
plt.ylabel('Millimeters')
```

```
plt.legend()
plt.show()
```




```
YEARS = np.linspace(1951, 2024, 74)
plt.figure(figsize=(14, 6))
plt.plot(YEARS, EUprcp, 'k-o', label='PRCP')
PRCP_ltl(EU, 1951, 2024, 'green')
plt.title('Eureka Yearly Total Precipitation (1951-2024)')
plt.xlabel('Year')
plt.ylabel('Millimeters')
plt.legend()
plt.show()
```



Sources:

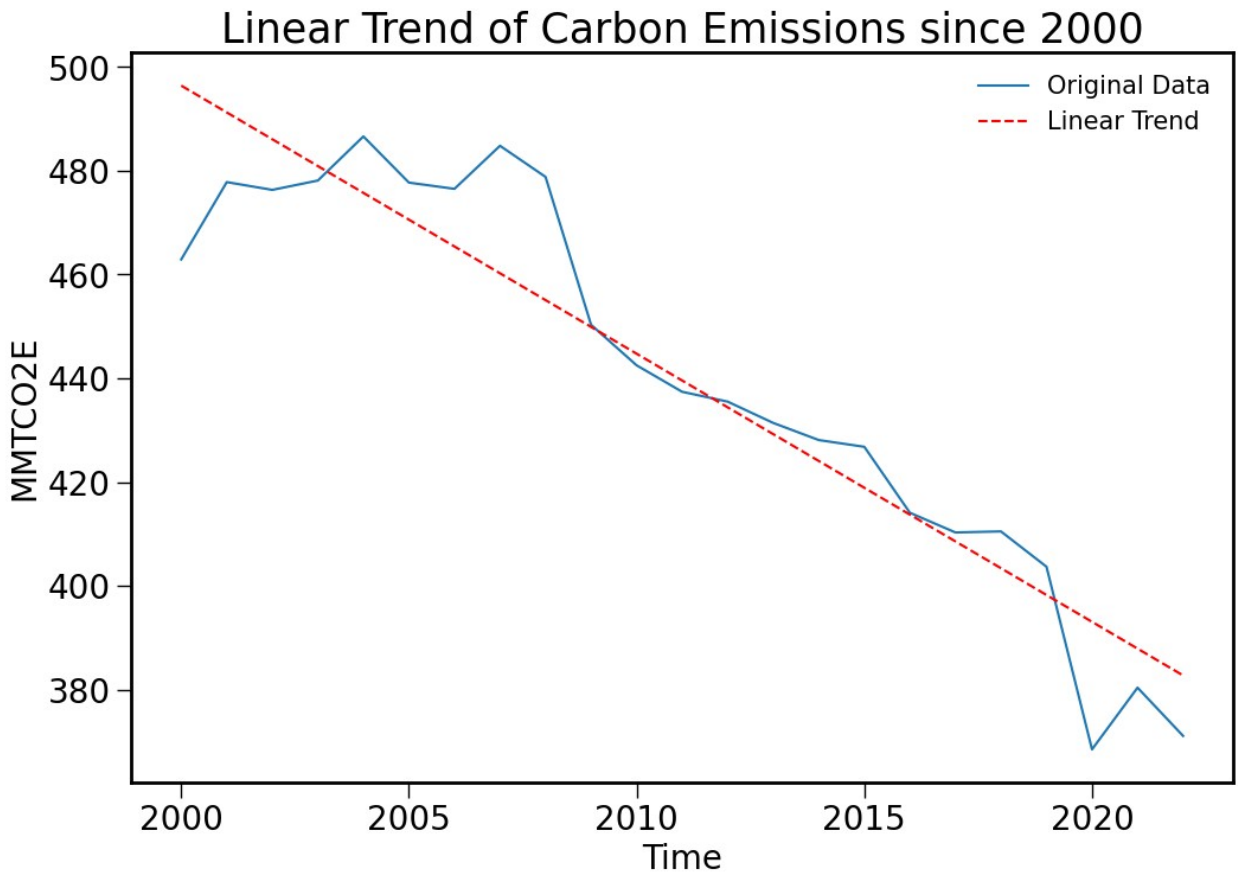
California Air Resources Board, "Current California GHG Emission Inventory Data 2000–2022," 2024. [Online]. Available: https://ww2.arb.ca.gov/ghg-inventory-data

```
c2023 = datas + '2023.txt'
C2023 = pd.read_csv(c2023, sep='\s+', header=None)
c2024 =  datas + '2024.txt'
C2024 = pd.read_csv(c2024, sep='\s+', header=None)

Car2023 = C2023.iloc[1:, 1:].astype(float)
total_2023 = np.nansum(Car2023.to_numpy())
Car2024 = C2024.iloc[1:, 1:].astype(float)
total_2024 = np.nansum(Car2024.to_numpy())
print(f"Total emissions for 2023: {total_2023}")
print(f"Total emissions for 2024: {total_2024}")
```

```
Total emissions for 2023: 17.773000000000003
Total emissions for 2024: 18.56
```

```
Carbon = np.array([462.9, 477.8, 476.3, 478.1, 486.6, 477.7, 476.5,
484.8, 478.8, 450.3, 442.5, 437.4, 435.5, 431.4, 428.1, 426.8, 414.1,
410.3, 410.5, 403.7, 368.5, 380.4, 371.1])

x = np.linspace(2000,2022,23)

slope, intercept = np.polyfit(x, Carbon, 1)

trend_line = slope * x + intercept

print(f"Slope: {slope:.3f}, Intercept: {intercept:.3f}")

plt.plot(x, Carbon, label='Original Data')
plt.plot(x, trend_line, color='red', linestyle='--', label='Linear
Trend')
plt.xlabel('Time')
plt.ylabel('MMTCO2E')
plt.legend()
plt.title('Linear Trend of Carbon Emissions since 2000')
plt.show()
```

```
Slope: -5.165, Intercept: 10826.822
```


Linear Trend of Carbon Emissions since 2000
Original Data
Linear Trend
MMTCO2E
500
480
460
440
420
400
380
2000
2005
2010
2015
2020
Time